Project 3

Comparison of Deep Learning Architectures

Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks

1. Introduction

Deep learning has produced a range of neural network architectures, each designed around the shape and nature of the data it processes. This report studies three foundational architectures — the Feedforward Neural Network (FNN), the Convolutional Neural Network (CNN), and the Recurrent Neural Network (RNN) — and compares them in terms of their internal structure, the type of input data they are best suited for, and their common real-world applications.

2. Basic Architecture

2.1Feedforward Neural Network (FNN)

A Feedforward Neural Network is the simplest type of artificial neural network. It consists of an input layer, one or more hidden layers, and an output layer, with neurons in each layer fully connected to the neurons in the next layer. Data moves in a single direction — forward from input to output — with no loops or feedback connections. Each connection has its own weight, which is adjusted during training through backpropagation.

- Layers: Input layer → one or more hidden layers → output layer.
- Connections: Fully connected (dense); every neuron connects to every neuron in the next layer.
- Memory: None — each input is processed independently of previous inputs.

Training: Backpropagation with gradient descent to minimize prediction error.

2.2Convolutional Neural Network (CNN)

A Convolutional Neural Network is designed to process grid-structured data such as images. Instead of connecting every neuron to every input value, a CNN applies small filters (kernels) that slide across the input to detect local features such as edges, textures, and shapes. These convolutional layers are typically followed by pooling layers, which reduce the spatial size of the data, and finally by fully connected layers that perform the final classification or regression.

- Layers: Convolutional layers → activation (e.g., ReLU) → pooling layers → fully connected layers.
- Connections: Locally connected with shared weights (filters), rather than fully connected throughout.
- Memory: None across samples, but preserves spatial relationships within a single input.
- Key idea: Parameter sharing and local receptive fields drastically reduce the number of parameters compared to a fully connected network.

2.3Recurrent Neural Network (RNN)

A Recurrent Neural Network is built to handle sequential data, where the order of inputs carries meaning. Unlike FNNs and CNNs, an RNN contains recurrent (looped) connections: the output of a neuron at one time step is fed back as part of the input for the next time step. This gives the network a form of memory, allowing it to use information from earlier in the sequence when processing later elements. Variants such as LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit) were developed to help RNNs retain information over longer sequences.

- Layers: A repeating recurrent unit applied at each time step, carrying a hidden state forward.
- Connections: Include feedback loops in addition to forward connections.
- Memory: Maintains a hidden state that captures information from previous time steps.
- Key idea: Parameter sharing across time steps allows the network to generalize across sequences of varying length.

3.Suitable Applications

3.1Feedforward Neural Networks

- Basic classification and regression problems on structured/tabular data.
- Credit scoring and risk assessment.
- Simple pattern recognition tasks where inputs are independent of one another.

3.2 Convolutional Neural Networks

- Image classification (e.g., identifying objects in photographs).
- Object detection and facial recognition.
- Medical image analysis, such as detecting tumors in scans.
- Video analysis, when combined with additional temporal processing.

3.3Recurrent Neural Networks

- Natural language processing tasks such as language modeling and text generation.
- Machine translation between languages.
- Speech recognition and audio processing.
- Time-series forecasting, such as stock price or weather prediction.

4.Comparison of Architectures

The table below compares the three architectures based on their structure, the type of input data they process, and their typical real-world applications.

Aspect	Feedforward Neural Network (FNN)	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Structure	Fully connected layers arranged in a single forward direction; no cycles or memory of previous inputs.	Convolutional layers with shared filters/kernels, followed by pooling layers and fully connected layers at the end.	Neurons with recurrent (looped) connections that feed the output of a step back as input to the next step, giving the network memory.
Input Data	Fixed-size, structured/tabular numerical data.	Grid-like data such as images or video frames, where spatial relationships between pixels matter.	Sequential or time-series data such as text, speech, or sensor readings, where order matters.
Information Flow	One-directional, from input layer to output layer.	One-directional, but with local receptive fields that scan across the input.	Cyclical; information persists across time steps via hidden state.
Parameter Sharing	None — every connection has its own independent weight.	Yes — the same filter weights are reused across the whole input, reducing parameters.	Yes — the same weights are reused across every time step.
Typical Applications	Basic classification/regression, credit scoring, simple pattern recognition.	Image classification, object detection, facial recognition, medical image analysis.	Language modeling, machine translation, speech recognition, time-series forecasting.
Key Strength	Simplicity; good baseline for structured, non-sequential data.	Excellent at capturing spatial hierarchies and local patterns in images.	Excellent at capturing temporal dependencies and context across a sequence.

Aspect	Feedforward Neural Network (FNN)	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Key Limitation	Cannot capture spatial or temporal patterns; treats all inputs independently.	Not naturally suited to sequential data with variable-length dependencies.	Can be slow to train and may struggle with very long-term dependencies (mitigated by LSTM/GRU variants).

5. Conclusion

Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks each address a different kind of data and problem. FNNs offer a simple, general-purpose architecture best suited to structured data with no spatial or temporal pattern. CNNs are optimized for grid-like data such as images, using convolutional filters to capture spatial hierarchies efficiently. RNNs are designed for sequential data, using recurrent connections and hidden states to capture temporal dependencies. Understanding these structural differences makes it possible to choose the right architecture for a given task, and in practice many modern systems combine these building blocks — for example, using a CNN to extract features from video frames before passing them to an RNN for temporal analysis.